

Movie and Employee Review Analysis

A Data Mining and Text Analysis Project

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RapidMiner, Excel

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This report includes About the Datasets, Section 2, Section 3, and Conclusion sections. It is about the IMDb_Top_Movies (Dataset 1) and Tata_Motors_Employee_Review (Dataset 2) datasets. Dataset 1 consists of data for IMDb's top movies. Dataset 2 consists of data for employees at Tata Motors (an Indian multinational auto manufacturing company), and displays their reviews of the company.

About the Datasets:

Dataset 1:

Dataset Name: IMDb_Top_Movies

Introduction: The dataset consists of both numerical and categorical data. There are around 10,000 (9,849 to be exact) movies represented in the dataset. There are 11 columns used to organize the data in the dataset, which are, Title, Year, Runtime, Certificate, Genre, Director, Stars, Rating, Metascore, Votes, and Gross. You will learn more about these dataset attributes in the Data Dictionary portion of the Section 2 section. The data is well-structured.

Source of Data: The source of the data is the website, Kaggle, and it seems to be uploaded by Ashutosh Devpura. It was last updated 4 months ago, so the dataset likely represents IMDb's top movies data from 4 months ago. As mentioned on the source's site, the data presented in the dataset was extracted from the IMDb website.

Source Link: <https://www.kaggle.com/datasets/ashutoshdevpura/imdb-top-10000-movies-updated-august-2023/>

Dataset 2:

Dataset Name: Tata_Motors_Employee_Review

Introduction: The dataset consists of both numerical and categorical data. There are around 11,000 (10,752 to be exact) Tata Motors employee reviews given in the dataset. There are 14 columns used to organize the data in the dataset, which are, Title, Place, Job Type, Department, Date, Overall Rating, Work Life Balance, Skill Development, Salary and Benefits, Job Security, Career Growth, Work Satisfaction, Likes, and Dislikes. You will learn more about these dataset attributes in the Data Dictionary portion of the Section 2 section. The data is well-structured.

Source of Data: The source of the data is the website, Kaggle, and it seems to be uploaded by Manish Kumar. It was last updated 3 months ago, so the dataset likely represents the Tata Motors employee reviews from 3 months ago. As mentioned on the

source's site, the data presented in the dataset was extracted from Ambition Box, which is a platform where employees basically give reviews of their companies.

Source Link: <https://www.kaggle.com/datasets/manishkr1754/tata-motors-employee-reviews-dataset>

Section 2:

Dataset 1:

Dataset: IMDb_Top_Movies

Domain of dataset: The domain in which the dataset would be placed in would be Movies or Entertainment.

Data Dictionary:

Attributes of the Dataset:

Title – The name of the movie,

Year – The release year of the movie,

Runtime – The total duration of the movie (in minutes),

Certificate – The age certification (rating) of the movie,

Genre – The genre (category) of the movie,

Director – The director of the movie,

Stars – The star cast displayed in the movie,

Rating – The rating of the movie,

Metascore – The metascore rating based on the critical movie reviews,

Votes – The total quantity of user ratings or votes given to the movie on IMDb,

Gross – The total domestic box office gross earnings of the movie (in millions).

Data Cleaning:

The Metascore column had many missing values. The Metascore column was also not used to answer any of the Analysis Goals, so I removed it from the dataset.

That was basically the only data-related correction made, and everything else was set up well.

The other modification made to the dataset was a change to its layout. I formatted the dataset as a table and styled it differently to make the dataset more readable. The rows were colored, with one row being light gray and the other being kept the same with no fill at all. By doing so, each individual row was set apart from the other. I also filled the first row of the dataset, containing the data attributes, with dark gray, and set the text color to be white and bold, so that the heading could be easily recognized.

Dataset 2:

Dataset: Tata_Motors_Employee_Review

Domain of dataset: The domain in which the dataset would be placed in would be Automotive Manufacturing.

Data Dictionary:Attributes of the Dataset:

Title – The title of the employee's job

Place – The location of the company,

Job Type – The type of employee (full-time, part-time employee),

Department – The certain department within the company where the employee works,

Date – The date of review,

Overall Rating – Overall rating for the company (out of 5),

Work Life Balance – The company rating based on work-life balance (out of 5),

Skill Development – The company rating based the opportunities for skill improvement (out of 5),

Salary and Benefits – The company rating based on employee satisfaction with salary and benefits (out of 5),

Job Security – The company rating based on the job security at the company (out of 5),

Career Growth – The company rating based on career-opportunities (out of 5),

Work Satisfaction – The company rating based on employee satisfaction with work (out of 5),

Likes – The pros of the job/company,

Dislikes – The cons of the job/company.

Data Cleaning:

The dataset had 10,752 rows, which was more than the 10,000 rows requirement for the assignment. I removed the last 752 rows from the dataset, so that there would be a total of 10,000 rows.

That was basically the only data-related correction made, and everything else was set up well.

Just like with Dataset 1, the other modification made to Dataset 2 was a change to its layout. I formatted the dataset as a table and styled it differently to make the dataset more readable. The rows were colored, with one row being light gray and the other being kept the same with no fill at all. By doing so, each individual row was set apart from the other. I also filled the first row of the dataset, containing the data attributes, with dark gray, and set the text color to be white and bold, so that the heading could be easily recognized.

Used the HLOOKUP function to find the movie title with the greatest gross earning. Entered “title” as the *lookup_value*, and cells A1 to A9850 (the entire title column) as the *table_range*. Used the MATCH function containing the MAX function as the *lookup_value*, to find the row of the cell with the greatest gross earning. Within the MAX function, entered the range J1 to J9850 (the entire gross column), and then completed the MATCH function by reentering the cells J1 to J9850 as the *lookup_array*, and 0 to get an exact match. The MATCH function served as the *row_index* of the HLOOKUP function. At the end of the function, typed FALSE as the *range_lookup*, in order to get an exact match to the *lookup_value*.

The formula used to find the movie with the greatest gross earning is:

=HLOOKUP("title", A1:A9850, MATCH(MAX(J1:J9850), J1:J9850, 0), FALSE)

To get the rating of that movie, used the VLOOKUP function. Entered "Star Wars: Episode VII – The Force Awakens" (movie title with greatest gross earning) as the *lookup_value*. Selected cells A1 to J9850 (the whole table) and named the range as IMDb_Top_Movies_Data. Inserted IMDb_Top_Movies_Data as the *table_range*, entered 8 as the *column_index*, because the "rating" column is 8 away from the first column to the left, and entered FALSE to get an exact match. The formula used to get the rating of the movie with the greatest gross earning is:

=VLOOKUP("Star Wars: Episode VII – The Force Awakens",
IMDb_Top_Movies_Data, 8, FALSE)

Used the VLOOKUP function to get the gross earning of the movie. The formula is:

=VLOOKUP("Star Wars: Episode VII – The Force Awakens",
IMDb_Top_Movies_Data, 10, FALSE)

Inserted a PivotTable displaying the information.

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the fields, rating and gross, and added them to the Values area to display the sums of the rating and gross earnings for the movie. Selected the title field and added it to the Rows section. The Columns area contained the Values, which were the added rating and gross fields. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed. To select the specific movie, clicked on the dropdown arrow on the Movie Title heading, and selected the movie, "Star Wars: Episode VII – The Force Awakens".

To get the votes for that movie, used the VLOOKUP function. Entered “Knock Knock” (lowest-rated movie title) as the *lookup_value*. Inserted IMDb_Top_Movies_Data as the *table_range*, entered 9 as the *column_index*, because the “votes” column is 9 away from

the first column to the left, and entered FALSE to get an exact match. The formula used is:

=VLOOKUP("Knock Knock", IMDb_Top_Movies_Data, 9, FALSE)

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the fields rating and votes, and added them to the Values area. Selected the title field and added it to the Rows section. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed. To select the specific movie, clicked on the dropdown arrow on the Movie Title heading, and selected the movie, "Knock Knock".

Analysis Goal:

Goal Three: Which year were the most movies released in?

Results of the Analysis:

Answer to Question Three: The year with the most movie releases according to the dataset is 2018. There were 308 movies released in 2018.

Image 3: Q3 Chart & Table

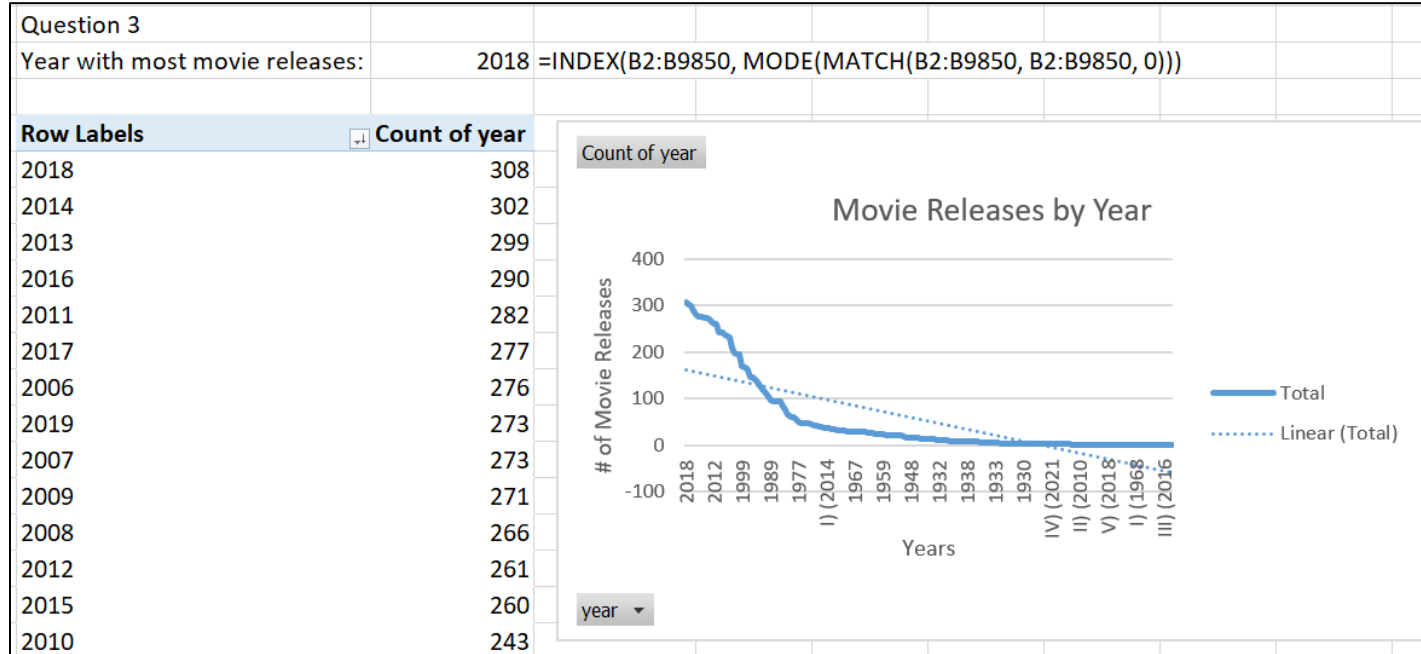


Image 3 displays the result for Question 3.

Calculation Steps:

Used the INDEX, MODE, and MATCH functions to find the year that appears the most.

The Year column contains the cells B2 to B9850. Within the INDEX function, entered the range B2:B9850 as the *array*, and the MODE function was inserted as the *row_num*.

The MATCH function served as *number1* within the MODE function. Entered B2:B9850 as both the *lookup_value* and *lookup_array* in the MATCH function, and 0 as the *match_type* to get an exact match. The formula used is:

=INDEX(B2:B9850, MODE(MATCH(B2:B9850, B2:B9850, 0)))

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the year field, and added it to the Values and Rows areas to display the counts of the years, and the year itself. Adjusted the years to be in descending order based on the counts. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed.

PivotChart – Selected a cell in the PivotTable and clicked on Insert and then PivotChart. Selected the Line chart, which was suitable for displaying the data in the PivotTable, and pressed OK. By doing so, the PivotChart was created. Next, clicked on the plus sign at the top right corner of the chart and selected the Axis Titles, Chart Title, and Trend Line chart elements in addition to Axes, Gridlines, and Legend, which were already selected. Titled the chart as Movie Releases by Year. The x-axis represented the Years and the y-axis represented the Number of Movie Releases.

Analysis Goal:

Goal Four: Which year had the greatest average domestic box office movie gross earnings?

Results of the Analysis:

Answer to Question Four: According to the dataset, 1996 had the greatest average domestic box office movie gross earnings, with gross earnings averaging \$242 (in millions). 2022 was second in domestic gross earnings with an average of \$156 (in millions) gross earnings.

Image 4: Q4 Chart & Table

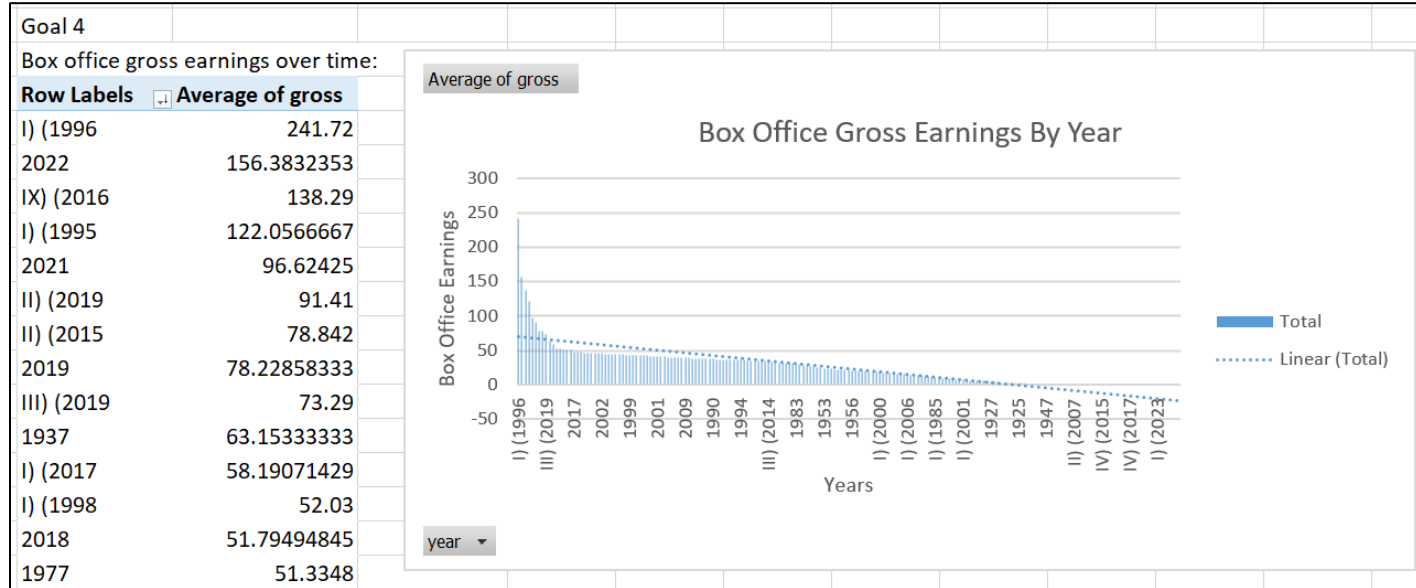


Image 4 displays the result for Question 4.

Calculation Steps:

Inserted a PivotTable and PivotChart to display the average box office gross earnings by year.

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the year field, and added it to the Rows areas to display the years, and added the gross field to the Values area to display the average gross earnings. Adjusted the years to be in descending order based on the average gross earnings. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed.

PivotChart – Selected a cell in the PivotTable and clicked on Insert and then PivotChart. Selected the Bar chart, which was suitable for displaying the data in the PivotTable, and pressed OK. By doing so, the PivotChart was created. Next, clicked on the plus sign at the top right corner of the chart and selected the Axis Titles, Chart Title, and Trend Line chart elements in addition to Axes, Gridlines, and Legend, which were already selected. Titled the chart as Box Office Gross Earnings by Year. The x-axis represented the Years and the y-axis represented the Box Office Earnings.

Analysis Goal:

Goal Five: Which movie genre had the highest average rating?

Results of the Analysis:

Answer to Question Five: According to the dataset, the combination of Comedy, Family, and Mystery movie genres, and the combination of Adventure and Western movie genres, had the highest average rating, with both having an average rating of 8.8. It basically means that people usually like to watch movies of these genres.

Image 5: Q5 Chart & Table

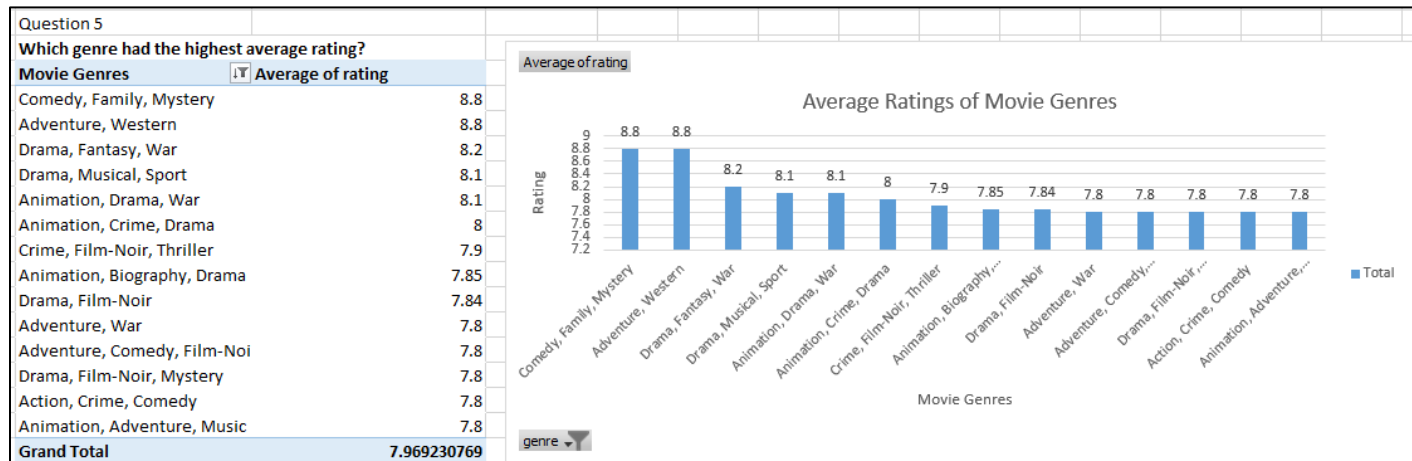


Image 5 displays the result for Question 5.

Calculation Steps:

Inserted a PivotTable and PivotChart to display the average ratings of movie genres.

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the genre field, and added it to the Rows areas to display the movie genres, and added the rating field to the Values area to display the average ratings. Adjusted the movie genres to be in descending order based on the average ratings. To shorten the list of movie genres, clicked on the dropdown arrow on the Movie Genres heading, and selected the top 10 items to be displayed based on the average

rating. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed.

PivotChart – Selected a cell in the PivotTable and clicked on Insert and then PivotChart. Selected the Bar chart, which was suitable for displaying the data in the PivotTable, and pressed OK. By doing so, the PivotChart was created. Next, clicked on the plus sign at the top right corner of the chart and selected the Axis Titles, Chart Title, and Data Labels chart elements in addition to Axes, Gridlines, and Legend, which were already selected. Titled the chart as Average Ratings of Movie Genres. The x-axis represented the Movie Genres and the y-axis represented the Rating.

Analysis Goal:

Goal Six: Which movie has the longest runtime?

Results of the Analysis:

Answer to Question Six: According to the dataset, the movie, Satantango has the longest runtime with 439 minutes (That's around 7 hours of movie runtime). It is interesting to find out that a movie had a runtime of around 7 hours, which is a lot for a movie.

Image 6: Q6 Table

Question 6									
Movie with the longest runtime:	Satantango	=HLOOKUP("title", A1:A9850, MATCH(MAX(C1:C9850),C1:C9850,0), FALSE)							
Runtime:	439	=VLOOKUP("Satantango", IMDb_Top_Movies_Data, 3,FALSE)							
Movie Title		Movie Runtime							
Satantango		439							

Image 6 displays the result for Question 6.

Calculation Steps:

Used the HLOOKUP function to find the movie title with the longest runtime. Entered “title” as the *lookup_value*, and cells A1 to A9850 (the entire title column) as the

table_range. Used the MATCH function containing the MAX function as the *lookup_value*, to find the row of the cell with the longest movie runtime. Within the MAX function, entered the range C1 to C9850 (the entire runtime column), and then completed the MATCH function by reentering the cells C1 to C9850 as the *lookup_array*, and 0 to get an exact match. The MATCH function served as the *row_index* of the HLOOKUP function. At the end of the function, typed FALSE as the *range_lookup*, in order to get an exact match to the *lookup_value*. The formula used:

=HLOOKUP("title", A1:A9850, MATCH(MAX(C1:C9850), C1:C9850, 0), FALSE)

To get the runtime of that movie, used the VLOOKUP function. Entered "Satantango" (movie title with the longest runtime) as the *lookup_value*. Inserted IMDb_Top_Movies_Data as the *table_range*, entered 3 as the *column_index*, because the "runtime" column is 3 away from the first column to the left, and entered FALSE to get an exact match. The formula used is:

=VLOOKUP("Satantango", IMDb_Top_Movies_Data, 3, FALSE)

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the runtime field and added it to the Values area. Selected the title field and added it to the Rows section. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed. To select the specific movie, clicked on the dropdown arrow on the Movie Title heading, and selected the movie, "Satantango".

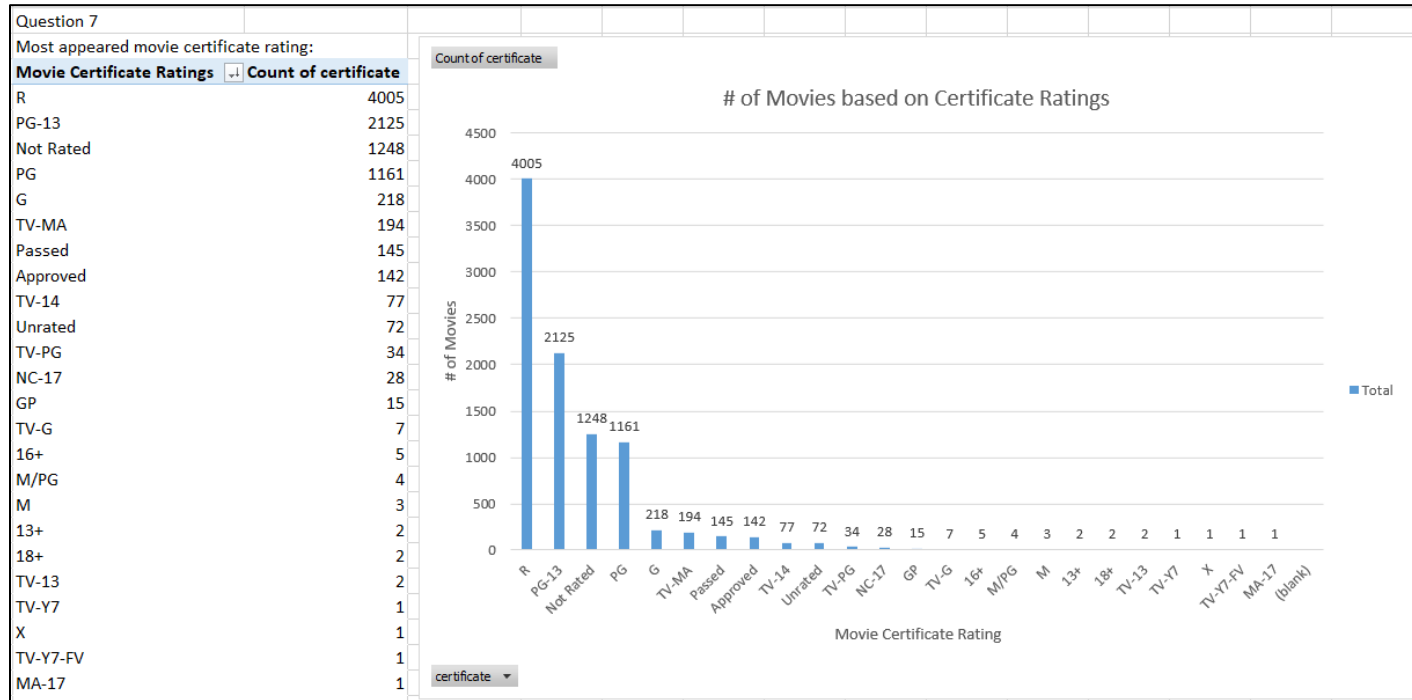
Analysis Goal:

Goal Seven: What is the most common certificate rating for the movies displayed in the dataset?

Results of the Analysis:

Answer to Question Seven: The most-appeared movie certificate rating according to the dataset, is the R movie certificate rating, with it applying to 4,005 movies.

Image 7: Q7 Chart & Table



The Pie chart displayed in Image 7: Q7 Chart 2, also nicely indicates that the majority of movies were rated R according to the dataset.

Image 7: Q7 Chart 2

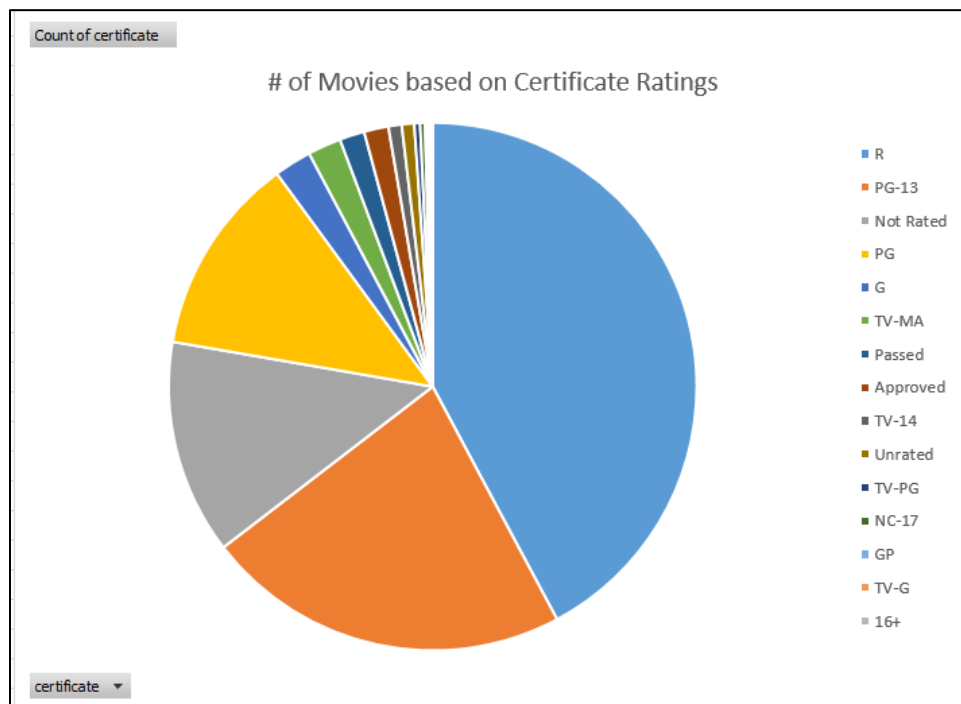


Image 7 displays the result for Question 7.

Calculation Steps:

Inserted a PivotTable and 2 PivotCharts to display the number of movies based on movie certificate ratings.

PivotTable – Selected a cell in the IMDb_Top_Movies_Data dataset and clicked on Insert and then PivotTable. Selected the certificate field, and added it to the Values and Rows areas to display the names and counts of the movie certificate ratings. Adjusted the movie certificate ratings to be in descending order based on the counts. After selecting the fields and arranging them in the preferred areas, the PivotTable was formed.

Bar Chart – Selected a cell in the PivotTable and clicked on Insert and then PivotChart. Selected the Bar chart, which was suitable for displaying the data in the PivotTable, and pressed OK. By doing so, the PivotChart was created. Next, clicked on the plus sign at the top right corner of the chart and selected the Axis Titles, Chart Title, and Data Labels chart elements in addition to Axes, Gridlines, and Legend, which were already selected. Titled the chart as # of Movies based on Certificate Ratings. The x-axis represented the Movie Certificate Rating and the y-axis represented the # of Movies.

Pie Chart - Selected a cell in the PivotTable and clicked on Insert and then PivotChart. Selected the Pie chart, which in addition to the Bar chart, was also suitable for displaying the data in the PivotTable, and pressed OK. By doing so, the PivotChart was created. Next, clicked on the plus sign at the top right corner of the chart and selected the Chart Title element in addition to the Legend, which was already selected. The legend displayed to the right of the pie chart, uses different colors to represent the different movie certificate ratings. Titled the chart as # of Movies based on Certificate Ratings.

Section 4:

Dataset 2:

Analysis Goal:

Goal: What did most employees not like about working at Tata Motors? (What are the most mentioned cons?)

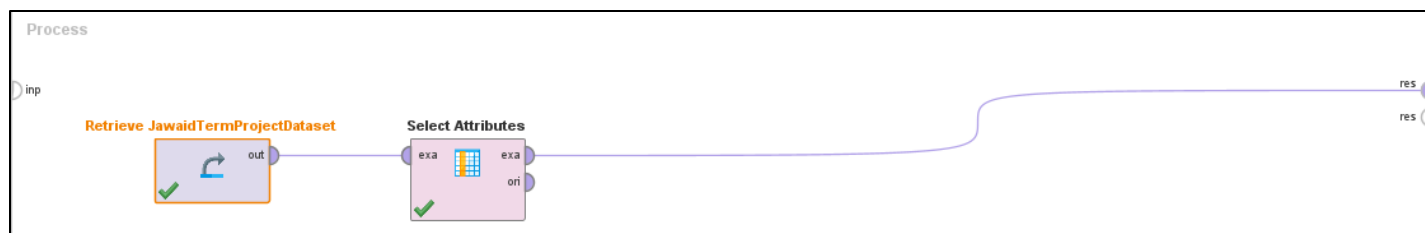
Results of the Analysis:

Answer to Question: Most employees were not satisfied with their work-life balance and career growth opportunities while working at Tata Motors.

Calculation Steps:

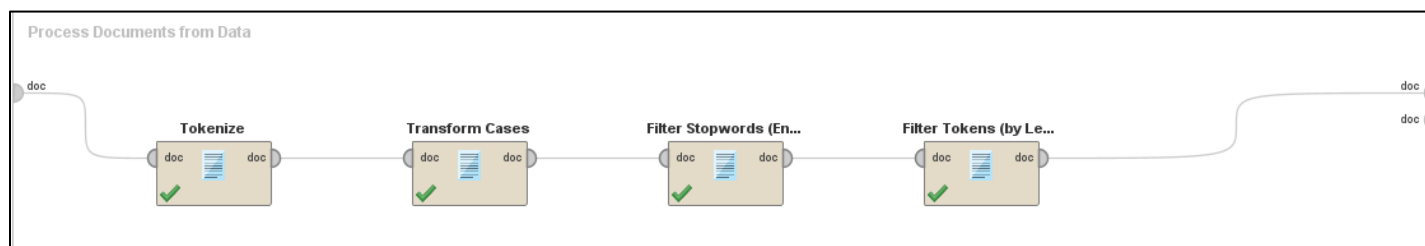
1. **Importing Dataset to RapidMiner:** Opened a blank new page on RapidMiner. Under the Repository section, clicked on Import Data and selected the downloaded Tata_Motors_Employee_Review file. Selected the default options and stored the file in a local repository as JawaidTermProjectDataset.
2. **Data Editing:** Under the Repository section, right-clicked on the JawaidTermProjectDataset file and selected Edit. In the Data Editor section, Right-clicked on the “Dislikes” column, selected Modify Attribute, and changed the Attribute Type from polynomial to text so that the data could be worked with in RapidMiner.
3. **Select Attributes Operator:** To add the dataset to a new model, dragged the JawaidTermProjectDataset dataset under the Repository section to the Process window. That adds a retrieve operator which connects to the dataset. Connected the output port with the result port through the use of a spline, making a data stream. In the Operators section, searched for Select Attributes, under the Selection folder. Added the Select Attributes operator to the data mining stream. The Select Attributes operator is used to decrease the number of attributes being focused on. Image 8 displays these steps in RapidMiner.

Image 8:



4. **Process Documents from Data Operator:** Then searched for Process Documents from Data, under the Text-Processing folder, and dragged the operator to the Process window, adding it to the data stream. Selected the Select Attributes operator and in the Parameters section, set the attribute filter type to “one attribute” and selected the “Dislikes” attribute. Next, clicked on the Process Documents from Data operator and under the Parameters section, set “vector creation” to Binary Term Occurrences, checked the “keep text” box, and set “absolute” as the prune method. Entered 2 for “prune below absolute” which means that the words that appear in less than 2 documents are not taken into consideration. Entered 9999 for “prune above absolute” which means that words that are displayed in more than 9999 documents are ignored or not taken into consideration. Double-clicked the Process Documents from Data operator and added multiple textual processing operators within it.
5. **Tokenize Operator:** In the Operators section, searched for Tokenize under the Tokenization folder, and dragged the Tokenize operator to the Process window. Connected the doc port with the doc ports at both sides of the panel. Clicked on the operator and in the Parameters section, selected non-letters for the mode.
6. **Transform Case Operator:** Searched for Transform Cases in the Operators section under the Transformation folder and added the operator to the data stream. Clicked on the operators and under the Parameters section, selected lower case for “transform to”
7. **Filter Stopwords Operator:** Next, in the Operators section, searched for the Filter Stopwords (English) operator under the Filtering folder and added it to the data stream.
8. **Filter Tokens Operator:** Then searched for the Filter Tokens (by Length) operator under the Filtering folder in the Operators section, and added the operator to the data stream. Selected the operator and under the Parameters section, enter 4 for “min chars” and 99 for “max chars”. That means that a token must have a minimum of 4 and a maximum of 99 characters in order for it to be taken into consideration. Image 9 displays the added operators within the Process Documents from Data operator.

Image 9:

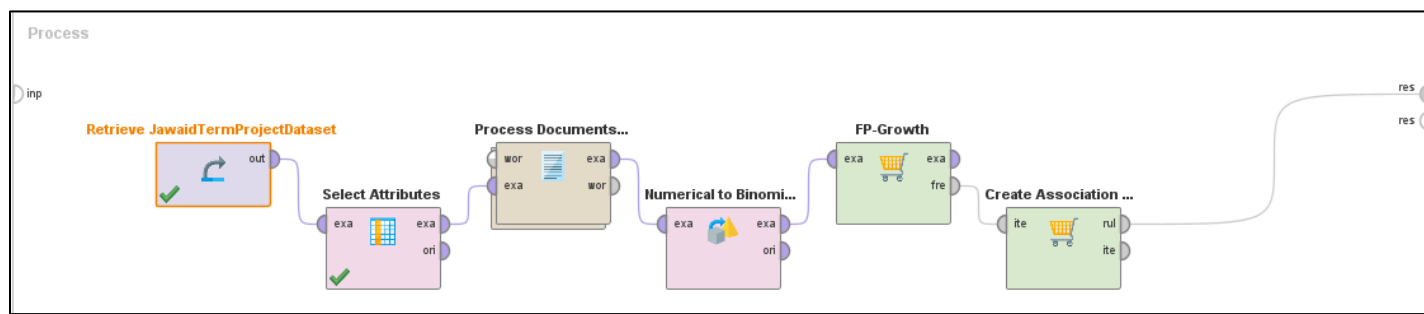


9. **Numerical to Binomial Operator:** Went back to the main Process window, and in the Operators section, searched for the Numerical to Binomial operator under the Types folder and added it to the data stream with the operator’s exa port

connected to the exa port of the Process Documents from Data operator. The Numerical to Binomial operator changes the attributes with a numeric data type to a binomial data type.

10. **FP-Growth Operator:** In the Operators section, searched for the FP-Growth operator under the Associations folder and dragged the operator to the Process window, adding it to the data stream. Clicked on the operator and set 0.05 as the “min support” and the “max items per itemset” had a 0, which means that there is no limit for the size of the itemsets. FP means Frequency Pattern, and the FP-Growth operator measures frequencies of the attributes and assists with determining the association rules.
11. **Create Association Rules Operator:** In the Operators section, searched for the Create Association Rules operator under the Associations folder and dragged it to the Process window. Connected the fre port on the FP-Growth operator to the ite port on the Create Association Rules operator, and connected the rul port of that operator to the res port at the right of the Process window. Selected the Create Association Rules operator and set the “min confidence” to 0.2, which means that the association rules have a minimum confidence of 0.2. Clicked the Run option, and Results view was displayed by default, showing the association rules for the “Dislikes” column. The Create Association Rules operator generates the association rules. Association rules display the connections between the attributes found in the dataset. Image 10 displays the completed main Process window.

Image 10:



Findings: After going through the steps mentioned in Section 4 of this report, clicked the Run option and got the association rules for the “Dislikes” column. In the Results tab, selected the Description section at the left, and a list of the association rules was displayed. The list is shown in Image 11. The answer to the question is represented by the last lines of the list, which have the greatest confidence. The last line of the list shows: [life, balance] → [work] (confidence: 0.978). In the “Dislikes” column, the words life and balance were shown to be strongly associated with the word, work, with a confidence

level of 98%. Career growth also had a high confidence of 84% shown by the line:

[career] → [growth] (confidence: 0.844). That tells us that most Tata Motors employees complained about the work-life balance along with the career growth opportunities they got while working at the company.

Image 11: Association Rules of “Dislikes” of working at Tata Motors

```

Association Rules
[tata] --> [good] (confidence: 0.202)
[motors] --> [good, tata] (confidence: 0.202)
[motors] --> [good] (confidence: 0.204)
[tata, motors] --> [good] (confidence: 0.206)
[work] --> [culture] (confidence: 0.207)
[growth] --> [salary] (confidence: 0.209)
[tata] --> [work] (confidence: 0.216)
[salary] --> [growth] (confidence: 0.222)
[motors] --> [work, tata] (confidence: 0.222)
[good] --> [company] (confidence: 0.222)
[motors] --> [work] (confidence: 0.224)
[tata, motors] --> [work] (confidence: 0.225)
[company] --> [work] (confidence: 0.232)
[work] --> [good] (confidence: 0.233)
[motors] --> [company, tata] (confidence: 0.245)
[motors] --> [company] (confidence: 0.246)
[tata, motors] --> [company] (confidence: 0.249)
[company] --> [tata] (confidence: 0.251)
[good] --> [work] (confidence: 0.251)
[tata] --> [company] (confidence: 0.262)
[company] --> [good] (confidence: 0.325)
[culture] --> [good] (confidence: 0.352)
[growth] --> [slow] (confidence: 0.356)
[growth] --> [career] (confidence: 0.395)
[environment] --> [good] (confidence: 0.573)
[culture] --> [work] (confidence: 0.591)
[environment] --> [work] (confidence: 0.652)
[company, tata] --> [motors] (confidence: 0.728)
[life] --> [work, balance] (confidence: 0.734)
[life] --> [balance] (confidence: 0.750)
[tata] --> [motors] (confidence: 0.768)
[good, tata] --> [motors] (confidence: 0.782)
[work, tata] --> [motors] (confidence: 0.803)
[slow] --> [growth] (confidence: 0.824)
[career] --> [growth] (confidence: 0.844)
[work, life] --> [balance] (confidence: 0.852)
[life] --> [work] (confidence: 0.861)
[balance] --> [work, life] (confidence: 0.913)
[balance] --> [life] (confidence: 0.934)
[balance] --> [work] (confidence: 0.949)
[work, balance] --> [life] (confidence: 0.962)
[life, balance] --> [work] (confidence: 0.978)

```

Conclusion:

By analyzing datasets through the use of RapidMiner and various Excel functions and data summarization/visualization tools like PivotTables and PivotCharts to answer questions about the datasets, I have gained a better understanding of how analysis is performed using Excel and RapidMiner. Specifically, about the dataset, IMDb_Top_Movies, I learned many interesting things, with one being that according to the dataset, 2018 had the most movie releases, however, it did not have the greatest average domestic box office movie gross earning. Overall, the analysis went well. The process of completing this assignment did provide some experience performing an analysis through the use of RapidMiner and multiple Excel functions and tools. Analyzing datasets through the use of RapidMiner and Excel functions and tools provides you with interesting information about the data.

References:

Create a PivotChart. Microsoft Support. (n.d.-a). <https://support.microsoft.com/en-us/office/create-a-pivotchart-c1b1e057-6990-4c38-b52b-8255538e7b1c?ui=en-us&rs=en-us&ad=us>

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